SR. NO.	PAGE NO.	TITLE	DATE	TEACHER'S SIGN & REMARKS
1.	27.7	Basic Networking in Linux & Window OS	4/jun	30/7/18
2.	27.7	Learning & assigning of IP address		30/7/18
3.	27.7	Study of different types of cables and crimping of cable with RJV5. connector.		30/7/18
4.	2.8	Design a simple topology and configure with one react, 2 router and PC's using cisco packet tracker.		
5.	3.8	SETUP and configure LAN		
6.	11.8	Customize switch network module using CPT.		
7.	17.8	Develop a customized piri command to test server connectivity		
8.	20.8	Anonymous FTP server using python		
9.	30.8	Implement packet sniffer and save socket in python.		
10.	24.8	Design a single calculator using XMLRPC.		
11.	1.09	Develop a program to create reverse TCP count the path.		

SR. NO.	PAGE NO.	TITLE	DATE	TEACHER'S SIGN / REMARKS
12.	01.09	Capture and analyze network packet using line share		
13.	25.9	Analyse different types of web Page using Urban		
14.	05.09	Discard line chars using numpy server on try hachine platform.		

Experiment:-1

LINUX COMMANDS.

a) pwd

def:-

Displays the current working directory. Helps identify your current location in the file system.

~~eg:-~~

O/P:- /home/student

b) ls.

def:-

displays the current working and lists files and directories in the current folder.

output:- file1.txt Documents
Downloads.

c) cd.

def:-

Changes the current working directory. Used to navigate between folders

O/P:- cd Documents (No output if successful)

d) mkdir

def:-

Creates a new directory with the given name. Useful for organizing files.

O/P:- mkdir Project (no output)

e) rmdir
Removes an empty directory. It cannot delete directories containing files.
O/P:- rmdir Project (No output)

f) touch
Creates a new empty file. Also updates the timestamp of an existing file
O/P :- touch notes.txt

g) cp
Copies files or diretories to another location. Use -r to copy directories.
O/P :- cp file1.txt
backup.txt

h) cat
Displays the content of the file. Can also combine multiple files.
O/P :- Hello World

i) whoami
Displays the username of the current logged-in user.
O/P :- student

j) ps:
Displays currently running processor. Help monitor active programs.
O/P :- PID TTY TIME CMD.

WINDOWS COMMAND

a) ipconfig

def:- View IP configuration

```
Windows IP Configuration

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::9549:f60a:e2fd:3d08%9
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

b) ping

def:- Test network connectivity

ping google.com

O/P:-

```
Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name
```

c) trace rt

def:- Trace the route to a destination

tracert google.com.

O/P:-

```
Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name
```

Experiment-2

LEARNING AND ASSIGNING IP ADDRESS

AIM:-

To study the concept of an IP address and assign a static IP address to computer for communication within a network.

REQUIREMENTS:-

Computer, Windows operating system, Wifi, Control Panel, Command prompt.

THEORY:-

What is an IP address?

An Internet Protocol (IP) address is an unique numerical address assigned to every device connected to a computer network. It enables devices to identify each other and exchange data over the network. An IP address works like a house address, where IP address helps data packets reach the correct devices on a network.

VERSIONS OF IP ADDRESS:-

* IPv4 (Internet Protocol Version 4)
* IPv6 (Internet Protocol Version 6)

TYPES OF IP ADDRESS:-

1. Static IP address,

A static IP address is manually assigned to a device by the user/network administrator. It remains the same until it is changed manually.

2. Dynamic IP address,

A dynamic IP address is automatically assigned by DHCP server whenever a device connects to the network.

PROCEDURE:-

Step 1:- Open Control Panel.

Step 2:- Select Network and Internet

Step 3:- Open Network and Sharing Center.

Step 4:- Click change adapter Settings

Step 5:- Right-click active network connection & Select Properties.

Step 6:- Select Internet Protocol Version 4 (TCP/IPV4)

Step 7:- Select Properties.

Step 8:- Choose Use the following IP address

Step 9:- Enter,

* IP addres: 192.168.1.100
* subnet Mask: 255.255.255.0
* Default gateway: 192.168.1.1.

Step 10:- Enter DNS Server.

* Preferred DNS: 8.8.8.8
* Alternate DNS: 8.8.4.4

Step 11:- Click OK and close all windows

Step 12:- Open Command prompt.

Step 13:- Type,

<//> cmd
ipconfig

Step 14:- Verify that the assigned IP address is displayed

OUTPUT:-

Command,

< / > cmd

ipconfig.

sample output:-

Windows IP configuration

Ethernet adapter Ethernet:

Connection-specific DNS suffix:

IPv4 Address ..... : 192.168.1.100

Subnet Mask ..... : 255.255.255.0

Default Gateway ..... : 198.168.0

RESULT:-

The concept of IP addressing was studied successfully. A static IP address was assigned to the computer, and the network configuration was verified using the ipconfig command. The assigned IP address enables successful communication.

Experiment - 3

Study of different types of network cables and crimping of cable with RJ45 connector.

AIM:-

To study the different types of network cables and crimp and ethernet cable using RJ45 connector.

TYPES OF NETWORK CABELS:-

1) Twisted pair cables:-
   → UTP (unshielded twisted pair)
   → STP (shielded twisted pair)
2) Coaxial cable
3) Optical fiber cache

RJ45 colour code - T568B (Default) order.

Pin	Colour	Pin	colour
1)	white - orange	1)	white - blue
2)	Orange	2)	Green
3)	white - green	3)	white - Brown
4)	Blue	4)	Brown

ORDER:-

White → orange → orange → white → green → blue → green → white → brown → brown.

INPUT:-

1) cat 5e/cat 6 Ethernet cable
2) RJ45 connector.
3) Cable tester.
4) Crimping table

Colour sequence Input:-

WO — O — WG — B — WB — G — WBr — Br

Output:-

The network cable was successfully crimped with an RJ45 connector using the to 68B colour standard, and the cable connection was verified using a cable tester.

30/10/25

EX:4

Design a simple topology and configure with one router, two switches and PC's using cisco packet tracking.

1. Build the topology:-

(i) Add end devices,

Navigate to end devices in the bottom-left palette place PC0 and PC1 on the left side (LAN1), place PC_2 and PC_3 on the right side (LAN2).

(ii) Add switches:-

Go to network devices > switches. Choose the cisco 2960 switch. Drag two switches onto the workshop.

(iii) Add a router,

Go to network devices > routers. chose the cisco 4331 (or) 2911 router and place it in the centre, above the switches.

(iv) Connect both switches:-

Router 0 using straight-through cables.

2. Configure PC's:-

PC	Ipaddress	Subnet Mask	Gateway.
PC0	192.168.10.10	255.255.255.0	192.168.10.1
PC_1	192.168.10.11	255.255.255.0	192.168.10.1
PC_2	192.168.20.10	255.255.255.0	192.168.20.1
PC_3	192.168.20.11	255.255.255.0	192.168.20.1

3. Configure Router:-

enable
configure terminal
interface Gigabit Ethernet 01010
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface Gigabit Ethernet 0/0/1
ip address 192.168.20.1 255.255.255.0
no shutdown
end
write memory.

4. Verify connectivity:-

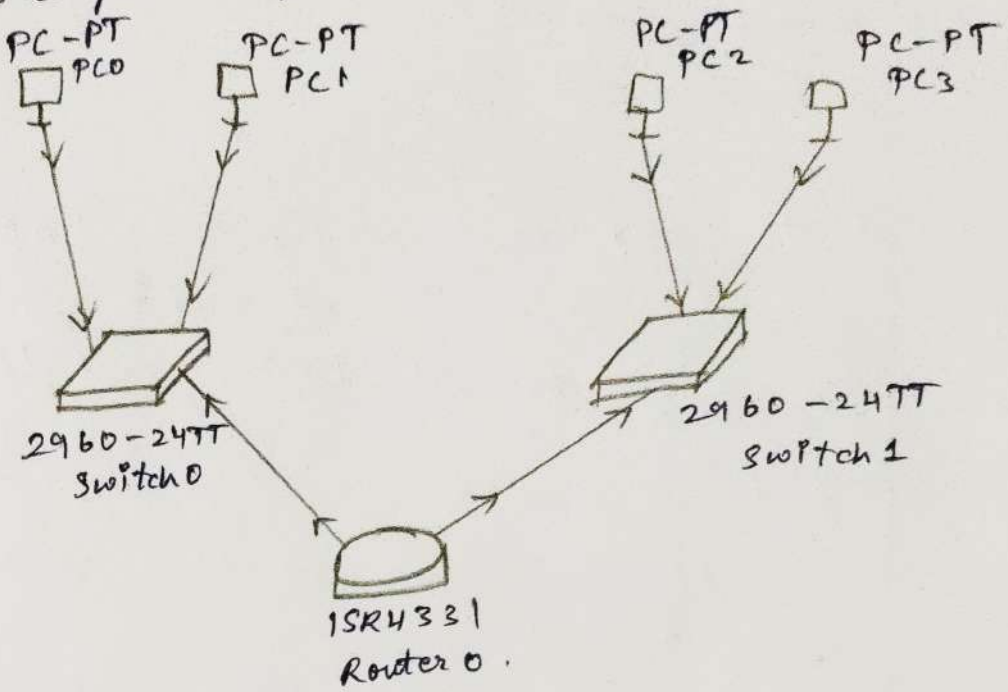
→ open PC → desktop → command prompt
→ Run

ping 192.168.10.1
ping 192.168.20.10

Result:-

The topology was successfully configured and communication between LAN A and LAN
was verified using ping.

Sample output :-



PC0

Physical Config Desktop Programming Attributes

IP address

Interface FastEthernet0

IP Configuration

○ DHCP ● Static

IPv4 address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

Exp :- 6

Customize switch with network modules using CISCO PACKET TRACER.

AIM :-

To customize a modular switch in cisco packet tracer by adding fast ethernet and gigabit fiber network modules and verifying the interfaces.

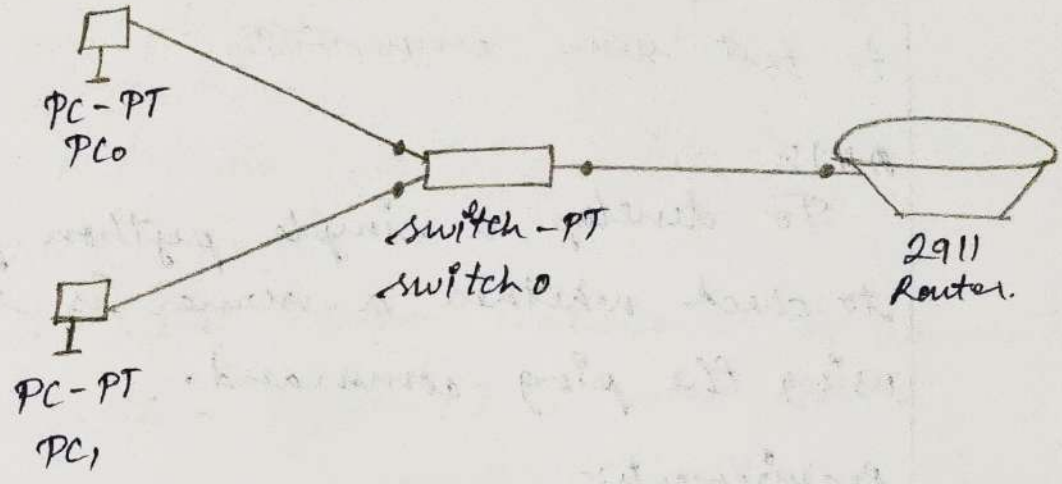
Procedure :-

1) Open cisco packet tracer and select network devices → switches → switch → PT empty
2) add 2 PCs and 1 router 2911 to workspace
3) Open the switch → physical tab → power off
4) Add modules,
   PT - switch - Nm - ICFE to slot 0, 1, 2
5) Configure IP address on the PCs and use ping to test connectivity.

Model reference :-

Connection	Module	Cable
PC0 → switch	Fast Ethernet 0/1	CFE (cable straight through)
PC1 → Switch	Fast Ethernet 0/1	CFE
Router → switch	Gigabit ethernet	CFE

Output:-



Result:-

The modular switch was successful customized with two fast ethernet ports and one gigabit fiber port & the interfaces were verified using CISCO IOSCLI.

EXP :- 7

Develop a customized ping command to test server connectivity.

AIM :-

To develop a simple python program to check whether a server is reachable using the ping command.

Requirements :-

Hardware :- * computer
* internet connection.

Software :- * Python
* VS code
* command prompt terminal

Theory :-

Ping is a network tool using to check whether a server is reachable. It uses ICMP to send requests to the destination and reciene replies.

* It takes a website on IP address as input.
* Sends 4 ping requests
* Displays ping results
* Shows whether server is reachable (or) unreachable.

Program:-

```
import subprocess
import platform
print ("=" * 50)
print ("Customized ping utility")
print ("=" * 40)
host = input ("Enter website or IP address")
if platform.system() == "windows":
        command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host].

try:
    result = subprocess.run (command,
                             capture_output = True)
print ("\n ping results \n")
print (result.stdout)
if result.returncode == 0:
    print ("server status: reachable")
else
    print ("server status: unreachable")

except Exception as e:
    print ("Error:", e)
```

Packets:-

Sent = 4
Received = 4
Lost = 0 (0% loss)
Approximate round trip times
Minimum = 27ms
Maximum = 30ms
Average = 28ms.

Result :-

Thus, the customized ping command was developed in python to test server connectivity.

EXP:-8

Anonymous FTP scanner using python

Aim:-

To develop a python program using ftplib to check whether an FTP server allows anonymous access.

Requirements:-

* Windows / Linux
* Python 3.0
* ftplib module
* Authorized FTP server.

Procedure:-

* Check python installation.
* Create a file named anonymous-ftp-scanner.py
Import FTP module.
Enter FTP server address
Connect to server and test anonymous login
Display accessible directories/files.

Program :-

```
from ftplib import FTP, all_errors
def anonymous_ftp_scan host, port = 2
print ("=", *40)
print ("Anonymous FTP scanner")
print ("=", *40)
ftp (TP/L)
try
  print ("[1] connecting to FTP server....
  ftp.connect (host, port, timeout = 10)
  print ("[+] FTP server is reachable")
  print ("[+] Server:", stp.get welcome())
  print ("/n [2] Testing : anonymous login ....")
  ftp.login ("anonymous", "anonymous@
                    example.com")
  print ("[+] Anonymous . Login successful")
  print ("/n. [3] Checking directory ...")
  files = ftp.nlst()
  for item in files.
      print ("", item)
  ftp.quit()
except all-errors as error
      print ("[-] Anonymous login failed")
print ("[!] error :", error)
anonymous-ftp-start (target).
```

SUCCESS OUTPUT:-

ANONYMOUS FTP SCANNER

Target : 127.0.0.1
Port : 21

[1] connecting to FTP server
[+] FTP server is reachable
[+] Server response: 220 FTP server ready

[2] Testing anonymous login...
[+] Anonymous login SUCCESSFUL
[!] The FTP server permits anonymous access

[3] Checking accessible directory
[+] Directory listing is available:
welcome.txt
public
readme.txt.

FAILED OUTPUT:-

[1] connecting to FTP server...
[+] FTP server is reachable.

[2] Testing anonymous login...
[-] Anonymous login failed or FTP
[!] Error : 530 login incorrect.

Result:-
FTP server is reachable & allows annonymous access.

Exper: 9.

Implement pocket sniffer socket implem

AIM:-

To design a basic sniffer my python that equals network.

PROCEDURE:-

1) Install python codded it to the system path::

2) Create a progit path and file named sniffer.

3) Write a program into file.

4) Open command prompt as administer

5) Navigate to the page fetch

6) Pin the program

7) Observe the output.

CODE:-

```
import stret
import stct
host = socket.gethen gen(socket.sethost nam)
sniffer = sochet.socket (socket.AF.INET)
print ("Pocket sniffer startup")
try:
  whiteat:
    stow-dade_de = sniffer.refuse(165536)
```

```
print (version " -ip [0] >> 4)
print (TTL: " pp [5])
print (" Protocol:", ip[6:3])
print ((" sum pp:") socket . int _ntoa (int [8]))
expect keyport Integrit:
    sniffer . iotk (socket . SIO _RCVALL . socht .
                                    RCVALL_OTT)
print ("sniffr stopper").
```

OUTPUT :-

Packet sniffer starter

Version 4
TTL : 128
Postant : 6
some IP : 192.168.1.0
Destination IP : 192 168.8.1.1
Packet Length : 60
sniffer stopped

RESULT :-

The ~~path~~ pigen successfully contains network packets and displays the basic IP level information.

Exp: 10

Design as simple calculate using XML-PC

AIM:-

To deny and implementation a render calculate.

PRE-REQUESTS:

Basic python pigony knowlegge indistatly of client state.

ALGORITHM / procedure:-

(i) Input simple XML PRC server

2) Define add substract, multiply, divisible function.

3) Centre the sqan on local best let.

4) Register all funten

5) state the action server fact.

SERVER.PY:-

```
for xmm.retry import single
  def add (a,b):
     return a,b
  def substract (a,b):
     return a-b
  def multiply(a,b):
     return a*b
  def division (a,b):
     if b==0
  return "distant zero not allowed"
  return: save-forever().
```

CLIENT.PY:

```
import xmlrpc.client
a = float(input("Enter first numb:"))
b = float(input("Enter second numb:"))
print("Addition:", proxy.settle(a,b))
print("Subtraction", proxy.subtract(a,b))
print("Division", proxy.divide(a,b))
```

OUTPUT:-

XML-RPC calculate sum num of pat8000...

RESULT:-

The XMLRPC remote calculeulate was successfully implemented and scuked. The client remotely perform date, add, substatt, multiply and divide.

Exp:-11

Develop a program to create recv TCP something the python socket.

AIM:-

TO creat listen Tcp server connect using python between distant or local host.

PROCEDURE:-

List & properties:-

1) Input socket module
2) set host as 127.0.0.1 and port as 5000
3) Create a TCP socket my AP_INFT Bu module - ST scan
4) Can the module the and socket.

CONNECTOR PROGRAM:-

1) set host as 127.0.0.1 and port as 5000.
2) Create TCP socket and connect to listen
3) send a text message and recess reply
4) Dispaly replay aand disc to socket

PROGRAM CODE:-

& Listener:

```
Inport socket
host = "127.0.0.1"
port = 5000
server = socket.socket (socket.AF_NET.
        socket_sock_stream) host(.)
```

```
print ("connect to listner")
clint (connect(host .port))
clint send ("Hellow form connect "encod)
reply = client . recursion (1024) decode()
print ("Listner replied :", reply)
client . close()
```

LISTNER OUTPUT:-

```
waiting for receive committer:-
common for (1127.00.1, 5407 & 1)
receiver : Hello from connector.
Connect : output
connecty to listner
Listner replied Hello from listner
```

RESULT:-

TCP reserver connector establish between listner and connector on local host and merge to enthanged the phgl - sohlng.

Exp - 12:

CAPTURING AND ANALYSING NETWORK PACKE USING WIRE SNACK.

AIM:-

To capture, filter and analyse network and prone the output.

Description with procedure:-

1. open restricted and select the on local Area, connector Interface
2. Configig computer to stop automatically after 100 pockets.
3. Stop the capital if necessary and save the capture packet.
4. Enter the sequence protocal filter is its filter base inspect the packet.
5. Repeat the process very folly files.

- TCP : Tcp
- UDP : Udp
- ARP : arp
- DNS : dns
- HTTP : Http
- IP : ip
- ICMP : icmp
- DHCP : dhcp

6. Inspect packet details, including source and distination address - protocol fields and packet impaired source the captains for faster references.

RESULT:-

Network packet were captured and endorsed successfully using cisco.
The required protocol filter were applied to identify TCP, UDP, ARP, DNS, HTTP, IP, ICMP, DHCP. packet.